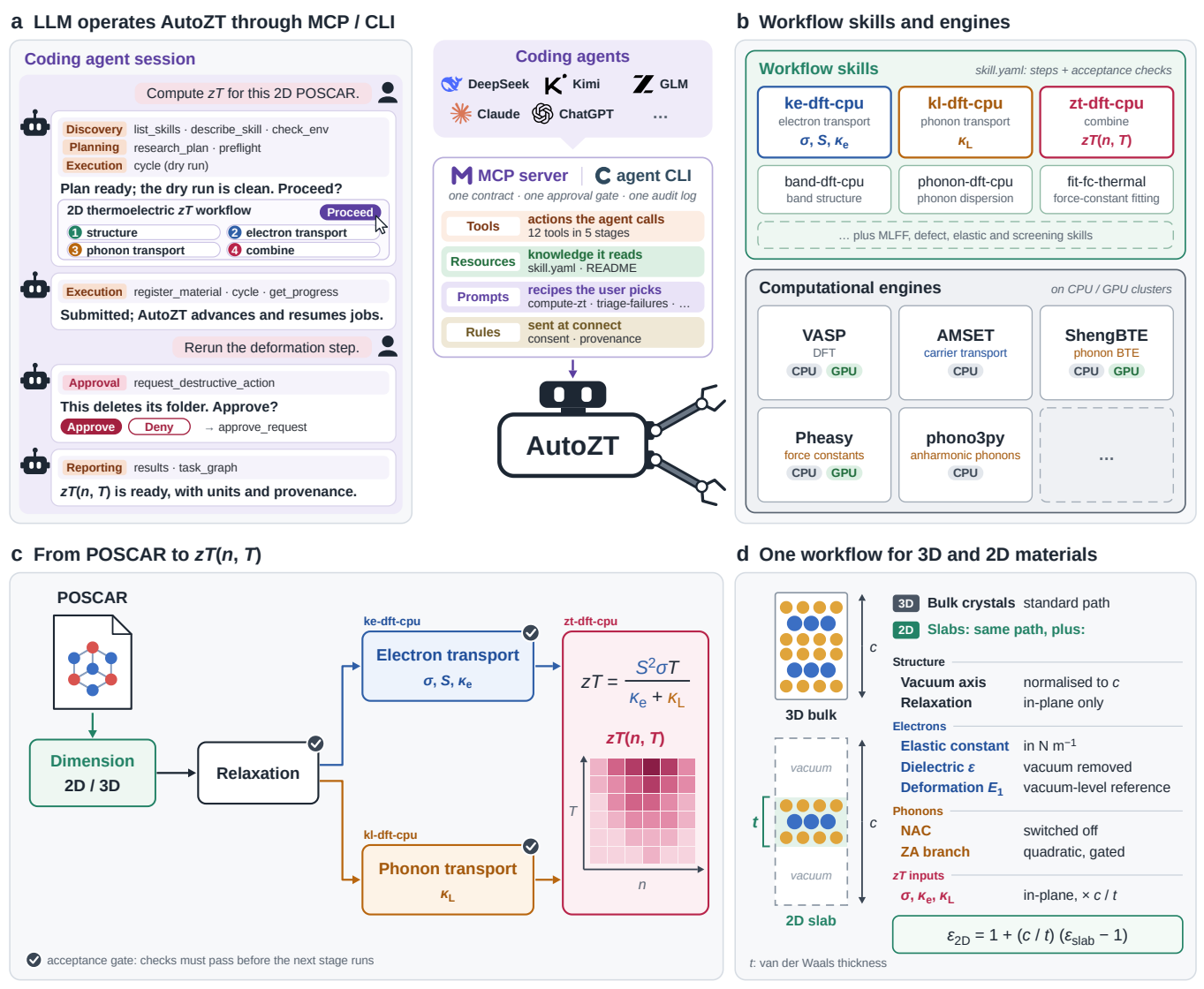

a LLM operates AutoZT through MCP / CLI
Coding agent session
Compute zT for this 2D POSCAR.
Discovery list_skills · describe_skill · check_env
Planning research_plan · preflight
Execution cycle (dry run)
Plan ready; the dry run is clean. Proceed?
2D thermoelectric zT workflow
Proceed
1 structure
2 electron transport
3 phonon transport
4 combine
Execution register_material · cycle · get_progress
Submitted; AutoZT advances and resumes jobs.
Rerun the deformation step.
Approval request_destructive_action
This deletes its folder. Approve?
Approve
Deny
→ approve_request
Reporting results · task_graph
zT(n, T) is ready, with units and provenance.
Coding agents
DeepSeek
Kimi
GLM
Claude
ChatGPT
...
MCP server
agent CLI
one contract · one approval gate · one audit log
Tools
actions the agent calls
12 tools in 5 stages
Resources
knowledge it reads
skill.yaml · README
Prompts
recipes the user picks
compute-zt · triage-failures · …
Rules
sent at connect
consent · provenance
AutoZT
b Workflow skills and engines
Workflow skills
skill.yaml: steps + acceptance checks
ke-dft-cpu
electron transport
σ, S, κe
kl-dft-cpu
phonon transport
κL
zt-dft-cpu
combine
zT(n, T)
band-dft-cpu
band structure
phonon-dft-cpu
phonon dispersion
fit-fc-thermal
force-constant fitting
… plus MLFF, defect, elastic and screening skills
Computational engines
on CPU / GPU clusters
VASP
DFT
CPU
GPU
AMSET
carrier transport
CPU
ShengBTE
phonon BTE
CPU
GPU
Pheasy
force constants
CPU
GPU
phono3py
anharmonic phonons
CPU
...
c From POSCAR to zT(n, T)
POSCAR
Dimension
2D / 3D
Relaxation
ke-dft-cpu
Electron transport
σ, S, κe
kl-dft-cpu
Phonon transport
κL
zt-dft-cpu
zT = S²σT / (κe + κL)
zT(n, T)
T
n
acceptance gate: checks must pass before the next stage runs
d One workflow for 3D and 2D materials
3D bulk
c
2D slab
vacuum
vacuum
t
c
3D Bulk crystals standard path
2D Slabs: same path, plus:
Structure
Vacuum axis normalised to c
Relaxation in-plane only
Electrons
Elastic constant in N m⁻¹
Dielectric ε vacuum removed
Deformation E1 vacuum-level reference
Phonons
NAC switched off
ZA branch quadratic, gated
zT inputs
σ, κe, κL in-plane, × c / t
ε2D = 1 + (c / t) (εslab − 1)
t: van der Waals thickness